

Gemini Slingshot - Complete Game Documentation

AI-Powered Bubble Shooter with Hand Gesture Control

For Global Game Jam 2026 - Theme: "MASK"

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1. Project Overview

What is Gemini Slingshot?

Gemini Slingshot is an innovative **AI-powered bubble shooter game** that combines classic bubble matching mechanics with cutting-edge technology:

- Hand Gesture Control:** Players use their webcam and natural hand movements (pinch-and-pull) instead of mouse/keyboard
- AI Strategic Co-pilot:** Google's Gemini 3 Flash AI analyzes the game board in real-time and suggests optimal moves
- Modern Web Tech:** Built with React 19, TypeScript, and Vite for a smooth 60fps experience

Key Features

Feature	Description
 Hand Tracking	MediaPipe Hands SDK tracks 21 hand landmarks for precise gesture recognition
 AI Analysis	Gemini 3 Flash Vision API processes screenshots and recommends strategic moves
 6-Color System	Strategic scoring with different point values per color (100-500 pts)
 Debug Panel	Real-time visibility into AI decision-making process
 Real-time Physics	Smooth ball trajectories with wall bouncing and collision detection

2. How It's Built

Technology Stack

FRONTEND STACK	
React 19.2	UI Components & State Management
TypeScript 5.8	Type Safety & Code Quality

Vite 6	Fast Build Tool & Dev Server
Tailwind CSS	Utility-First Styling
Lucide React	Icon Library
AI & VISION	
MediaPipe Hands	Real-time Hand Tracking (21 pts)
Google Gemini 3	Vision AI for Strategy Analysis
@google/genai	Official Gemini SDK

Project Structure

```
Gemini-Slingshot/
├─ App.tsx           # Root component wrapper
├─ index.tsx         # Application entry point
├─ index.html        # HTML template with MediaPipe CDN scripts
├─ types.ts          # TypeScript type definitions
├─ vite.config.ts    # Vite configuration
├─ package.json      # Dependencies & scripts
├─
├─ components/
│   └─ GeminiSlingshot.tsx # Main game component (~1000 lines)
│                           # - Game loop & rendering
│                           # - Physics simulation
│                           # - Hand tracking integration
│                           # - UI overlays
├─ services/
│   └─ geminiService.ts  # AI service layer (~200 lines)
│                           # - Gemini API integration
│                           # - Prompt construction
│                           # - Response parsing
│                           # - Fallback heuristics
├─ docs/
│   └─ ARCHITECTURE.md   # Technical architecture details
│   └─ API_REFERENCE.md  # Type definitions & functions
│   └─ DEPLOYMENT.md    # Hosting guide
│   └─ CONTRIBUTING.md   # Contribution guidelines
```

Core Type Definitions

```
// Game entities
interface Bubble {
  id: string;
  row: number;
  col: number;
  x: number;
  y: number;
```

```

    color: BubbleColor; // 'red' | 'blue' | 'green' | 'yellow' | 'purple' | 'orange'
    active: boolean;
}

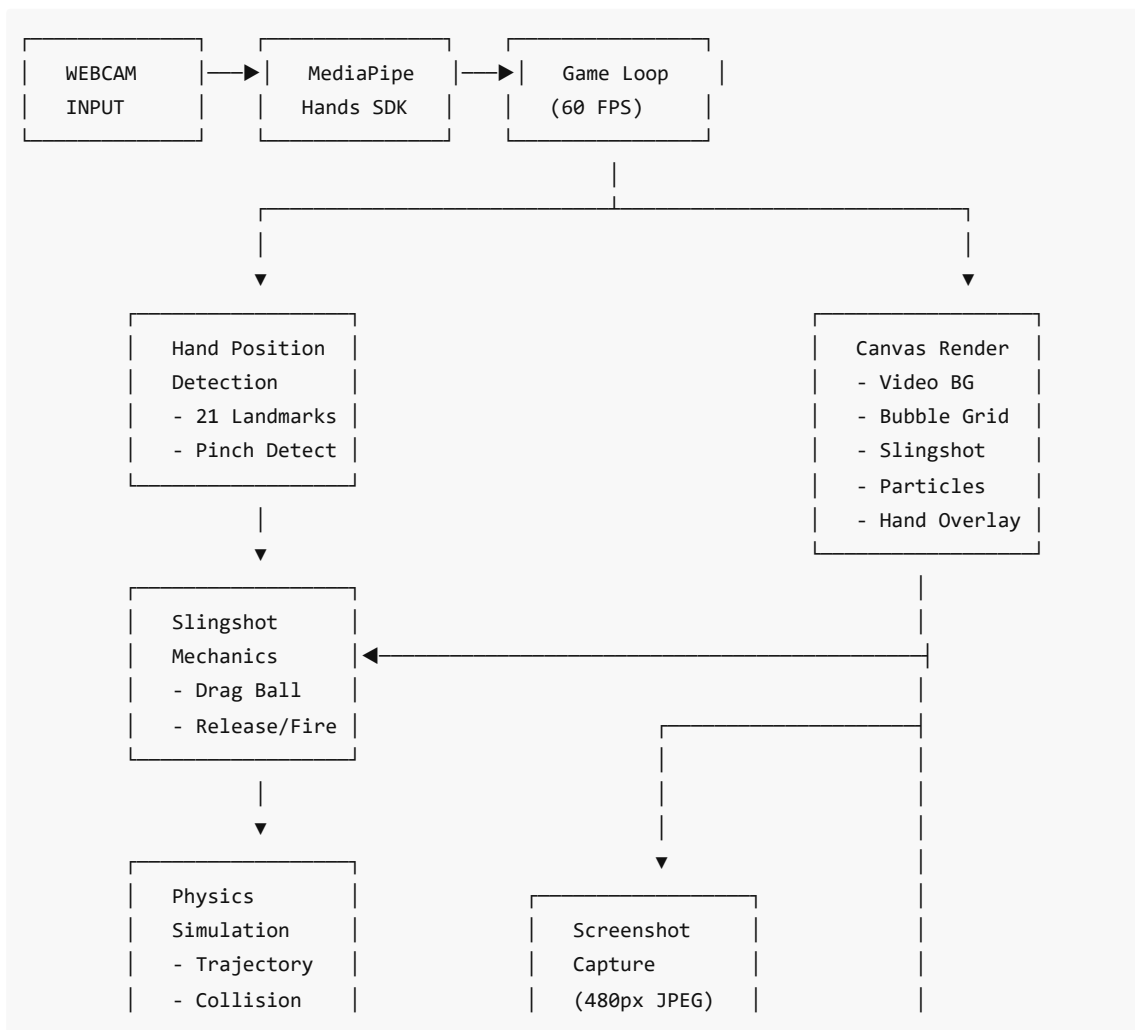
interface Point { x: number; y: number; }
interface Particle { x: number; y: number; vx: number; vy: number; life: number; color:
string; }

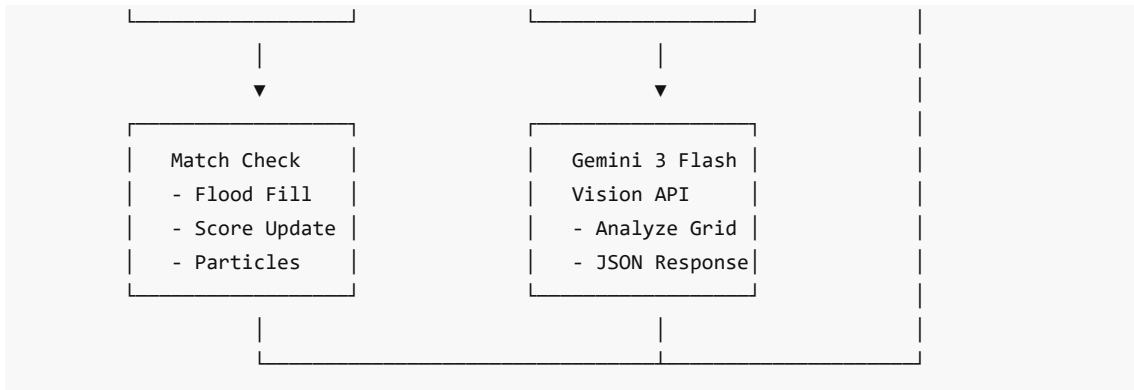
// AI response structure
interface StrategicHint {
    message: string;
    rationale?: string;
    targetRow?: number;
    targetCol?: number;
    recommendedColor?: BubbleColor;
}

```

3. How It Works

System Data Flow





Key Mechanics Explained

1. Hand Tracking & Pinch Detection

MediaPipe detects 21 hand landmarks. The game uses:

- Point 4: Thumb tip
- Point 8: Index finger tip

$\text{Pinch Distance} = \sqrt{(\text{thumb.x} - \text{index.x})^2 + (\text{thumb.y} - \text{index.y})^2}$

If Pinch Distance < 0.05 → User is "grabbing"

2. Slingshot Physics

When ball is released:

1. Calculate stretch distance from anchor
2. Apply velocity based on stretch (quadratic power curve)
3. Ball flies with:
 - Wall reflection (X-axis bounce)
 - Friction (0.998 per frame)
 - No gravity (space-like feel)
4. Collision detection uses substeps for fast-moving balls

3. Bubble Matching (Flood Fill Algorithm)

When ball snaps to grid:

1. Find all connected same-color bubbles
2. Use hex-grid neighbor detection (6 neighbors)
3. If cluster >= 3 bubbles:
 - Deactivate bubbles
 - Create explosion particles
 - Calculate score (base × combo multiplier)

4. AI Strategic Analysis

After each shot:

1. Capture canvas as 480px JPEG (60% quality)
2. Pre-calculate all reachable clusters for ALL colors
3. Send to Gemini with structured prompt:
 - Screenshot image

- Valid target options
 - Scoring rules
 - Board danger level
4. Parse JSON response for:
- Target coordinates
 - Recommended color
 - Strategic rationale
5. Display laser sight and update UI

Scoring System

Color	Points per Bubble	Strategic Value
 Red	100	Low - Common, easy targets
 Blue	150	Low-Medium
 Green	200	Medium
 Yellow	250	Medium-High
 Purple	300	High - Valuable targets
 Orange	500	Highest - Rare, priority target

Combo Multiplier: 1.5x when matching more than 3 bubbles at once

4. How to Play

Setup Requirements

- **Desktop computer** with webcam
- **Modern browser** (Chrome, Firefox, Edge)
- **Well-lit environment** for hand tracking
- **Node.js 18+** for development

Installation

```
# 1. Clone and install
cd Gemini-Slingshot
npm install

# 2. Configure API Key
# Create .env.local file with:
API_KEY=your_gemini_api_key_here

# 3. Start the game
npm run dev

# 4. Open http://localhost:3000
```

Game Controls

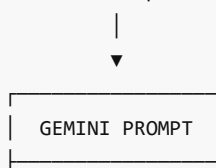
GAME CONTROLS
1. SHOW YOUR HAND 🖐️ Position your hand in view of the webcam
2. PINCH TO GRAB 🤏 Touch thumb to index finger near the slingshot ball
3. PULL BACK TO AIM 🎯 While pinching, move hand backward to stretch slingshot - Further back = more power - Band color changes to yellow when grabbed
4. RELEASE TO FIRE 🚀 Open your fingers to launch the ball
5. SELECT COLOR 🎨 Click color buttons at bottom to change ammo color - AI will auto-equip recommended color
6. FOLLOW AI HINTS 💡 Watch the laser sight line pointing to suggested target Read strategic advice in the right panel

Game Objectives

- 1. **Match 3+** bubbles of the same color to pop them
- 2. **Maximize score** by targeting high-value colors (Orange/Purple)
- 3. **Create chain reactions** by hitting high clusters to drop bubbles below
- 4. **Keep bubbles away** from the bottom of the screen
- 5. **Follow AI recommendations** for optimal strategic play

UI Elements

GAME CANVAS	DEBUG PANEL
🏆 SCORE 1,250 [BUBBLE GRID] o o o o o o o o o o o o o o o o o o o o o o o o o o o o o o o o	FLASH STRATEGY "Target the Orange cluster on right!"
[LASER SIGHT LINE]	VISION INPUT [Screenshot image]





6. New Feature Suggestions for "MASK" Theme

Global Game Jam 2026 Theme: "MASK"

The theme "MASK" can be interpreted in multiple ways: disguise, identity, protection, concealment, transformation. Here are innovative feature ideas to make Gemini Slingshot stand out:

TIER 1: HIGH-IMPACT FEATURES (Recommended for 48-hour jam)

1. Masked Bubbles

Concept: Some bubbles wear "masks" that hide their true color

Implementation:

- 20% of bubbles spawn as "masked" (gray/question mark appearance)
- True color revealed only when:
 - a) Hit directly by a ball
 - b) Adjacent to a match
 - c) AI special "unmask" ability used
- Strategic depth: Players must decide to probe unknowns or play safe
- Visual: Venetian mask overlay on bubbles with reveal animation

Why it's special: Adds mystery and strategic uncertainty - do you risk hitting an unknown?

2. Dual Identity Mode

Concept: Bubbles have TWO colors - one visible, one hidden beneath

Implementation:

- Each bubble has primary color (visible) and secondary color (masked)
- First match removes "surface mask" → reveals true color

- Second match of true color pops the bubble
- AI must predict hidden colors based on patterns
- Visual: Bubble appears to "crack" revealing inner color

Why it's special: Revolutionary twist on match-3 mechanics - every bubble is a two-layer puzzle

3. Player Mask Power-ups 🧐

Concept: Collectible masks that grant special abilities

Mask Types:

- 🦎 Chameleon Mask - Your ball takes the color of first bubble it touches
- 🔮 Oracle Mask - Reveals all masked bubbles for 10 seconds
- 👻 Phantom Mask - Ball passes through first row of bubbles
- 🛡️ Shield Mask - Prevents one bubble row from dropping
- 🎯 Precision Mask - Shows exact trajectory prediction

Implementation:

- Masks appear randomly in bubble grid
- Hitting a mask bubble activates it
- Single use, strategic timing matters
- Visual: Mask icon floats over player's hand

Why it's special: Power-up system that ties directly to theme

4. Emotion Masks AI Personality 🧐 😊 😈

Concept: The AI co-pilot wears different "masks" that change its personality

AI Personalities:

- 😊 Helpful Mask - Standard helpful hints (default)
- 😈 Trickster Mask - Occasionally gives misleading advice
- 🧐 Cryptic Mask - Gives hints as riddles
- 😡 Aggressive Mask - Focuses only on high-risk/high-reward plays
- 😌 Calm Mask - Focuses on safe, consistent plays

Implementation:

- Player can choose AI personality before game
- Each personality changes prompt behavior
- Visual: AI panel changes theme colors/icons

Why it's special: Meta-commentary on trusting AI - is the AI wearing a "mask"?

🎯 TIER 2: MEDIUM-EFFORT FEATURES

5. Face Mask Filter Reality 🧑 📺

Concept: Apply visual filters to player's face in webcam feed

Implementation:

- Use MediaPipe Face Detection alongside Hands
- Overlay themed masks on player's face:
 - Venetian masks

- Superhero masks
- Animal masks
- Mask changes based on score milestones
- Screenshot feature to share masked selfies

6. Unmasking Chain Reactions ✨

Concept: Special combo mechanic where matches "unmask" adjacent bubbles

Implementation:

- When 3+ bubbles match and pop:
 - All adjacent masked bubbles are revealed
 - If revealed colors also match → cascade!
- Creates unpredictable, exciting chain reactions
- Bonus points for "unmasking combos"

7. Hide & Seek Mode 🕵️

Concept: Alternating phases of "masked" and "revealed" gameplay

Implementation:

- 30 second cycles:
 - REVEAL PHASE: All bubbles visible, plan your strategy
 - MASK PHASE: Random bubbles get masked, must rely on memory
- Tests player memory and adaptation
- AI provides different hints in each phase

🌟 TIER 3: POLISH & DIFFERENTIATION

8. Theatrical Visual Theme 🎭

Concept: Complete visual overhaul with masquerade/theater aesthetic

Visual Elements:

- Background: Venice carnival scene / theater stage
- Bubbles: Styled as decorative masks
- Slingshot: Ornate golden theater prop
- Particles: Confetti and feathers
- UI: Art deco frames, theatrical typography
- Music: Mysterious carnival-style soundtrack

9. "Behind the Mask" Narrative 📖

Concept: Light story element that unfolds as you play

Implementation:

- Each level reveals part of a story about identity
 - AI co-pilot is a mysterious masked figure
 - Win condition: "Unmask the truth"
 - Creates emotional connection to theme
-

10. Multiplayer: Mask vs Unmask ✂️

Concept: Asymmetric 2-player mode

- Implementation:
- Player 1 (Masker): Tries to mask bubbles on opponent's board
 - Player 2 (Revealer): Tries to unmask and pop bubbles
 - Competitive twist on cooperative bubble shooter

📋 RECOMMENDED IMPLEMENTATION PRIORITY (48 Hours)

Priority	Feature	Time Est.	Impact
1	Masked Bubbles (random hidden colors)	4-6 hrs	★★★★★
2	Player Mask Power-ups (3 types)	4-6 hrs	★★★★★
3	Theatrical Visual Theme	8-10 hrs	★★★★★
4	Unmasking Chain Reactions	3-4 hrs	★★★★★
5	AI Personality Masks	4-5 hrs	★★★

🏆 What Makes This Stand Out at Global Game Jam

1. **Unique Tech Stack:** Hand gesture + AI is rare in jam games
2. **Theme Integration:** Masks aren't cosmetic - they're core mechanics
3. **AI Storytelling:** The "masked AI advisor" concept is meta and intriguing
4. **Accessible Yet Deep:** Easy to understand, hard to master
5. **Visual Polish:** Theater/masquerade aesthetic is striking
6. **Demo-Friendly:** Easy to show judges the hand tracking "wow factor"

Quick Reference Card

GEMINI SLINGSHOT QUICK REFERENCE CARD		
CONTROLS	SCORING	TIPS
 Pinch = Grab	 Red = 100	• Follow AI hints!
 Pull = Aim	 Blue = 150	• Target Orange first
 Open = Fire	 Green = 200	• Create avalanches
	 Yellow = 250	• Match 4+ for 1.5x
	 Purple = 300	• Watch the laser line
	 Orange = 500	
START: npm run dev → http://localhost:3000		
Built for Global Game Jam 2026		

Theme: "MASK"

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